Vulnerability Assessment and Penetration Testing (VAPT)

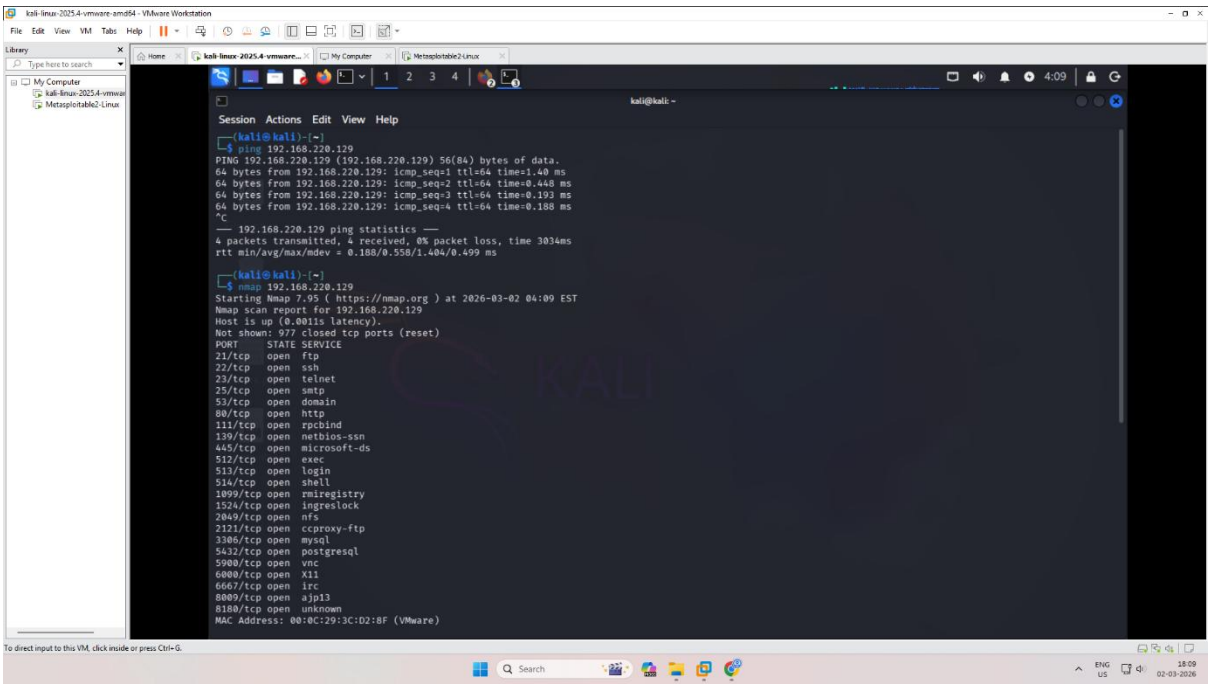
1. Introduction

This project demonstrates identifying vulnerabilities in a simulated lab environment and exploiting them using Kali Linux tools.

2. Tools Used

- Kali Linux
- Nmap
- Metasploit
- Burp Suite
- Metasploitable 2

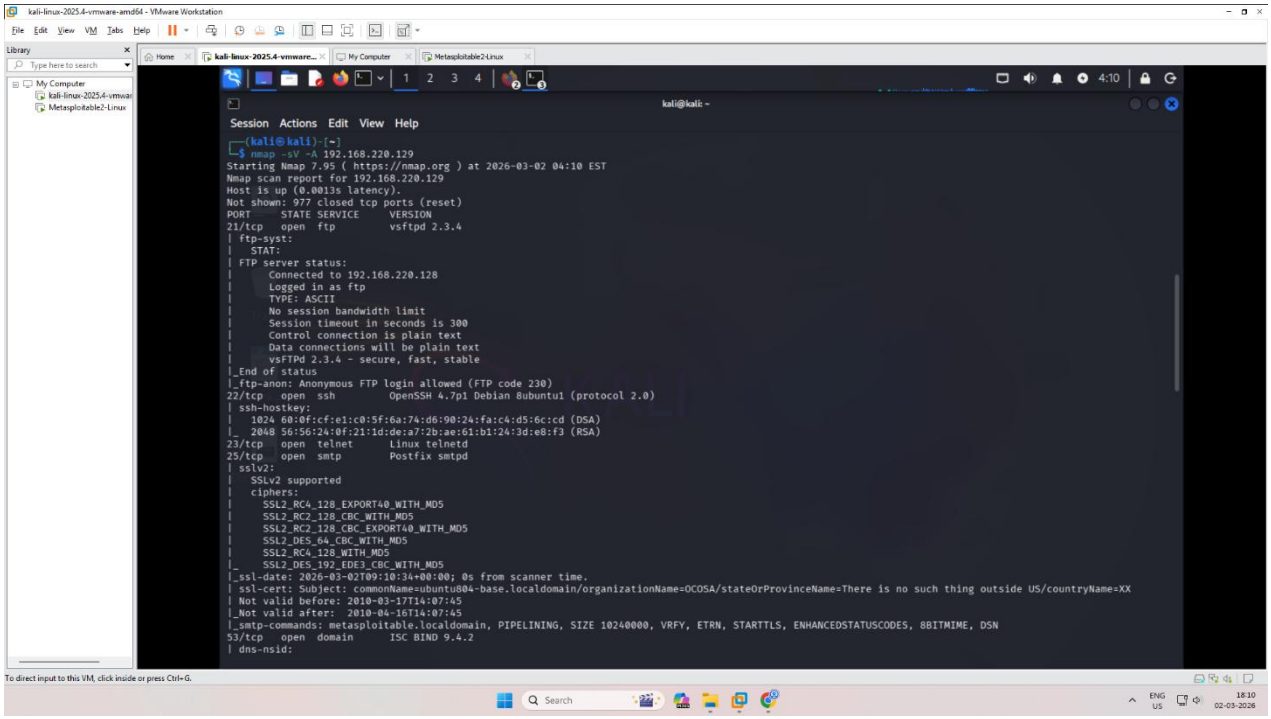
- **3. Methodology**
- **3.1 Reconnaissance**
- Target IP: 192.168.220.129
- Nmap scan results:

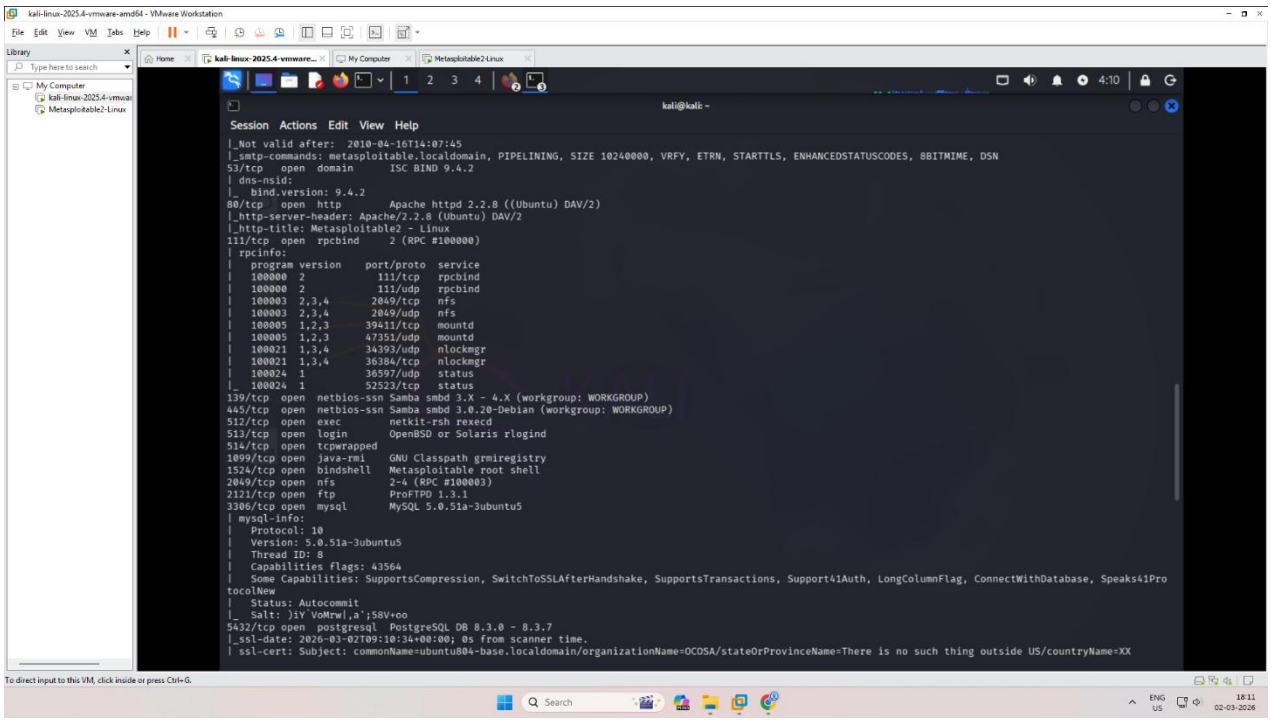


3.2 Vulnerability Assessment

Nmap vulnerability scripts identified:

- vsftpd backdoor vulnerability
- SMB vulnerability

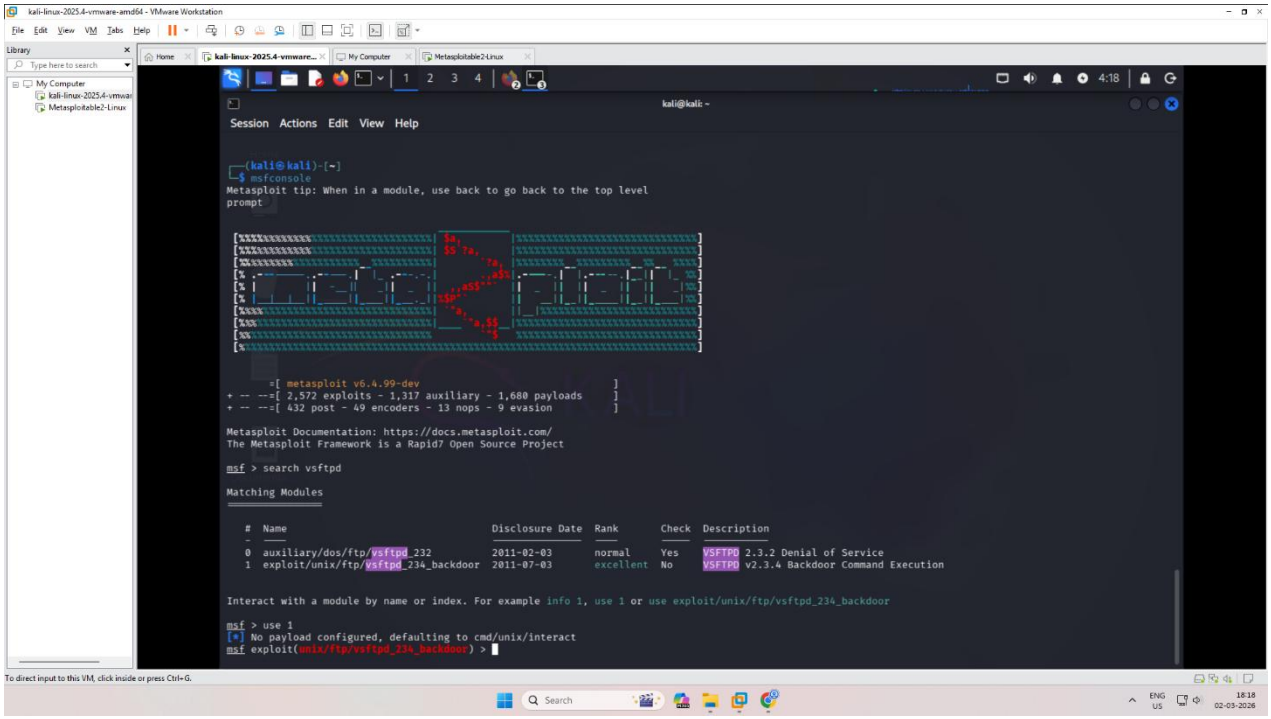


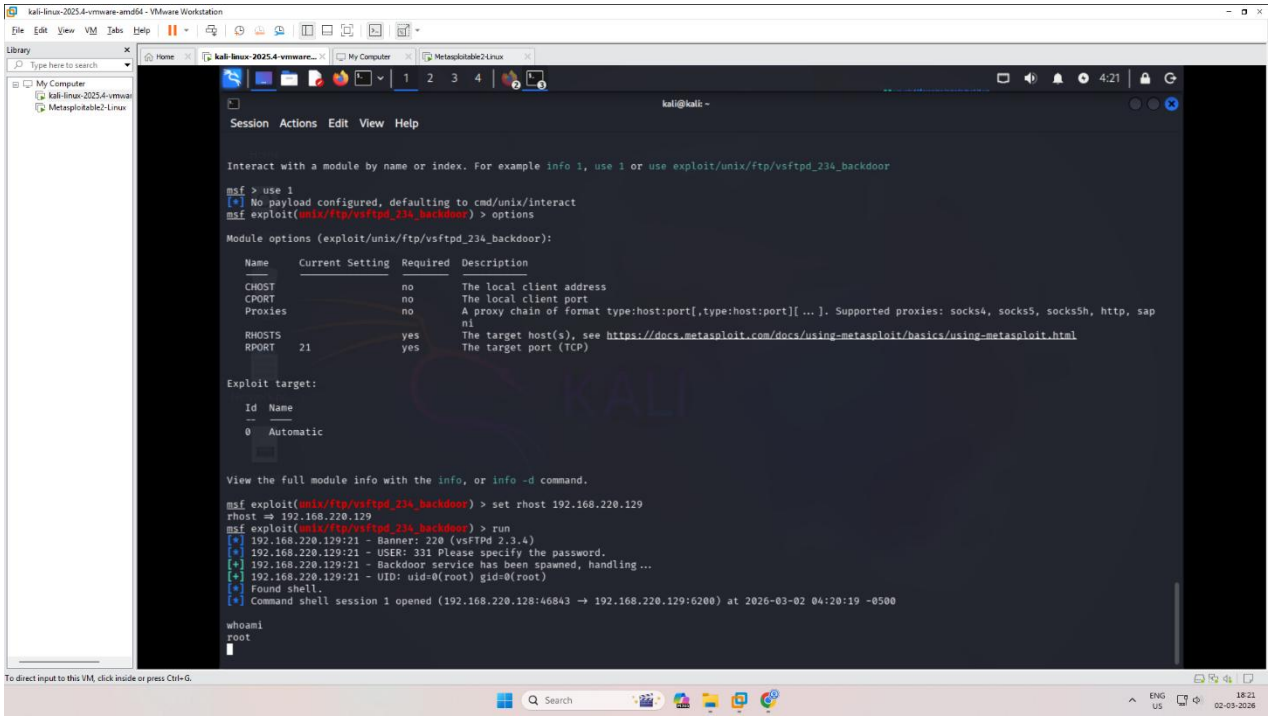


3.3 Exploitation

Metasploit was used to exploit vsftpd 2.3.4 vulnerability.

(Screenshot 4)





4. Findings

Vulnerability	Risk Level	Exploited	Status
vsftpd backdoor	High	Yes	Successful
Open ports	Medium	No	Observed

5. Conclusion

The VAPT process successfully identified and exploited vulnerabilities in a controlled lab environment, demonstrating the importance of patch management and secure configurations.